Appendix C.4. A Universal Canonical Improvement at Quadratic Order

For route-by-route flux comparisons, it is convenient to select the *same* improvement representative for all routes:

$$\boxed{J^{\mu}_{\rm can}[h^{\rm TT}] \;\equiv\; \frac{A_\star}{2}\, a^3 \Big(h^{\rm TT}_{ij}\, \nabla^{\mu} h^{\rm TT}_{ij} - \delta^{\mu}{}_{0}\, h^{\rm TT}_{ij}\, \dot{h}^{\rm TT}_{ij} \Big),} \tag{A36}$$

so that we *adopt the convention*

$$\delta^2 \mathcal{L}_X = A_\star \sqrt{-g}\, \mathrm{sgn}(\Sigma_\epsilon)\, I_T \;+\; \nabla_\mu J^{\mu}_{\rm can}, \qquad X \in \{ rank-one determinant route, \mathrm{CM}, \mathrm{CS}^+ \}. \tag{A37}$$

Different representatives differ by $\nabla_\nu X^{[\mu\nu]}$ and yield identical integrated fluxes under A4.

Check (FRW).

On spatially flat FRW in TT gauge,

$$\nabla_\mu J^{\mu}_{\rm can} = \frac{A_\star\, a^3}{2} \Big(\dot{h}^{\rm TT}_{ij} \dot{h}^{\rm TT}_{ij} - \frac{\partial_k h^{\rm TT}_{ij} \partial_k h^{\rm TT}_{ij}}{a^2} \Big) + \partial_\tau \big(\tfrac{3}{2} A_\star a^2 \dot{a}\, h^{\rm TT}_{ij} h^{\rm TT}_{ij} \big), \tag{A38}$$

i.e., the canonical reshuffling between TT kinetic/gradient bilinears plus a pure time boundary term that integrates to zero with A4 fall-offs.

Appendix C.5. Closed Forms for FRW and Weak Field

FRW.

For $ds^2 = a(\tau)^2(-d\tau^2 + d\mathbf{x}^2)$ and homogeneous $\epsilon(\tau)$,

$$J^0_{\rm can} = \frac{A_\star}{2}\, a^3\, h^{\rm TT}_{ij}\, \dot{h}^{\rm TT}_{ij}, \qquad J^i_{\rm can} = \frac{A_\star}{2}\, a\, h^{\rm TT}_{jk}\, \partial^i h^{\rm TT}_{jk}, \tag{A39}$$

which satisfies (A38).

Weak field (AF).

At $a \to 1$,

$$J^0_{\rm can} = \frac{A_\star}{2}\, h^{\rm TT}_{ij}\, \dot{h}^{\rm TT}_{ij}, \qquad J^i_{\rm can} = \frac{A_\star}{2}\, h^{\rm TT}_{jk}\, \partial^i h^{\rm TT}_{jk}. \tag{A40}$$

Appendix C.6. Flux-Ratio Identity and Finite-Domain Convergence

With the unified choice (A37), $\int_{\partial\mathcal{D}} d\Sigma_\mu\, J^{\mu}_{\rm ROD} = \int_{\partial\mathcal{D}} d\Sigma_\mu\, J^{\mu}_{\rm CM} = \int_{\partial\mathcal{D}} d\Sigma_\mu\, J^{\mu}_{\rm CS^+}$; hence,

$$\boxed{\mathcal{R}_{X/Y}[\partial\mathcal{D}] \equiv \frac{\displaystyle\int_{\partial\mathcal{D}} d\Sigma_\mu\, J^{\mu}_X}{\displaystyle\int_{\partial\mathcal{D}} d\Sigma_\mu\, J^{\mu}_Y} = 1, \qquad X, Y \in \{ rank-one determinant route, \mathrm{CM}, \mathrm{CS}^+ \}.} \tag{A41}$$

On finite FRW balls (or AF shells), residuals scale away with the radius R, agreeing with Section 5.

Summary of this Appendix.

At quadratic order and under A1–A5 plus C1, the rank-one determinant route, closed–metric, and PT-even CS/Nieh–Yan routes share the same bulk reduction $\delta^2\mathcal{L}_X = A_\star \sqrt{-g}\, \mathrm{sgn}(\Sigma_\epsilon)\, I_T \pmod{\nabla_\mu J^\mu}$, $A_\star = \lambda^2/8$. A single canonical improvement $J^{\mu}_{\rm can}$ (Equations (A36)–(A38)) is used for all routes and underlies the flux-ratio plots in Section 5. (Code: `fig_c2_coeff_compare.py`; `fig_flux_ratio.py`).